Victoria Park Academy Shenzhen

CIE IGCSE Computer Science (0478) -- YEAR 10 SEMESTER 2

COURSE OVERVIEW

Academic Year: 2025-2026 (Second Semester)

Course Code: CIE IGCSE Computer Science (0478)

Level: Year 10 (Final Semester Before IGCSE Exams)

Location: Victoria Park Academy, Shenzhen, China

Learner Profile: EAL (English as Additional Language), Mixed Ability

Lesson Duration: 40 minutes per period

SECTION 1: COURSE SUMMARY

COURSE DESCRIPTION (课程概述)

This is the FINAL and most critical semester of the IGCSE Computer Science course. By this point, students have completed all new content delivery across Year 9 (S1 and S2) and Year 10 Semester 1. Year 10 Semester 2 is dedicated entirely to INTENSIVE REVISION, MOCK EXAMINATION SERIES, GAP ANALYSIS, and FINAL EXAM PREPARATION.

The semester follows a structured pattern of: revise -- mock exam -- analyse -- target -- revise again -- mock exam -- final polish. This cyclic approach ensures maximum retention and confidence building before the actual IGCSE examinations.

WHAT STUDENTS HAVE ALREADY COMPLETED (学生已完成的内容)

Year 9 Semester 1 (Y9S1): All Paper 1 theory topics (Syllabus Topics 1-6)

- Data Representation (binary, hex, ASCII, images, sound)
- Data Transmission (serial/parallel, error detection, protocols)
- Hardware (CPU, memory, storage, I/O devices)
- Software (OS, utilities, translators, libre/open source)
- The Internet and Its Uses (IP addressing, domains, MAC, cookies, security)
- Automated and Emerging Technologies (robotics, AI, expert systems)

Year 9 Semester 2 (Y9S2): Paper 2 programming and algorithms (Syllabus Topics 7-8)

- Algorithm design (pseudocode, flowcharts)
- Programming fundamentals (variables, operators, selection, iteration, arrays)
- Subprograms (functions and procedures with parameters/return values)

Year 10 Semester 1 (Y10S1): Remaining topics + consolidation (Syllabus Topics 8-10)

- Databases (flat-file, relational, SQL queries)
- Boolean logic (logic gates, truth tables, combining gates, applications)

- Consolidation of all topics
- Mock Examination 1 (Weeks 17-18) -- Paper 1 and Paper 2

WHAT THIS SEMESTER COVERS (本学期内容)

Year 10 Semester 2 (Y10S2): EXAM READINESS AND MASTERY

- Unit 1: Paper 1 Intensive Revision (4 weeks)
- Unit 2: Paper 2 Intensive Revision (4 weeks)
- Unit 3: Mock Series 1 -- Mock 2 (Paper 1) + Mock 3 (Paper 2)
- Unit 4: Gap Analysis & Targeted Revision (4 weeks)
- Unit 5: Mock Series 2 -- Mock 4 (Paper 1) + Mock 5 (Paper 2)
- Unit 6: Final Preparation -- Revision Clinic & Exam Day Prep

SEMESTER GOALS (学期目标)

By the end of Y10S2, every student will be able to:

1. Recall and apply knowledge from all 10 syllabus topics with confidence
2. Answer Paper 1 short answer, structured, and extended questions accurately
3. Write pseudocode solutions to Paper 2 programming problems
4. Read, trace, and debug given pseudocode algorithms
5. Answer database/SQL and Boolean logic questions correctly
6. Complete a full 15-mark scenario question with structured reasoning
7. Manage examination time effectively (1.4 minutes per mark)
8. Identify their own weak areas and apply targeted revision strategies
9. Enter the IGCSE examination period with confidence and a clear revision plan

SECTION 2: HOW THIS SEMESTER CONNECTS TO PREVIOUS LEARNING

LEARNING JOURNEY MAP (学习旅程图)

Y9S1 (Foundation) >>>	Y9S2 (Skills) >>>	Y10S1 (Extension) >>>	Y10S2 (Mastery)
Learn theory	Learn programming	Learn DB/SQL/Boolean	REVISE EVERYTHING
Topics 1-6	Topics 7-8	Topics 8-10	Mock + Target + Polish
Build knowledge	Build skills	Fill gaps	Exam readiness
"Know WHAT"	"Know HOW"	"Know MORE"	"SHOW you know"

PEDAGOGICAL PROGRESSION (教学进展)

Y9S1 -- Knowledge Acquisition Phase

Students encountered concepts for the first time.

Focus: understanding definitions, memorising key facts.

Assessment: end-of-topic quizzes, vocabulary tests.

Y9S2 -- Skills Development Phase

Students applied knowledge through programming.

Focus: writing algorithms, coding solutions, problem-solving.

Assessment: coding assignments, pseudocode tasks.

Y10S1 -- Extension & Integration Phase

Students learned remaining topics and consolidated.

Focus: connecting ideas across topics, deeper understanding.

Assessment: Mock Exam 1 (first full exam experience).

Y10S2 -- Mastery & Examination Phase

Students demonstrate mastery through repeated practice.

Focus: exam technique, time management, error elimination.

Assessment: Mock Exams 2, 3, 4, 5 (progressive improvement).

WHY THIS STRUCTURE WORKS FOR VPA STUDENTS

- EAL learners benefit from repeated exposure to exam vocabulary
- Mixed-ability classes are supported through differentiated revision paths
- The mock-exam cycle builds familiarity and reduces exam anxiety
- Gap analysis ensures no student is left with unidentified weaknesses
- Final preparation week provides emotional as well as academic readiness

SECTION 3: 18-WEEK CURRICULUM MAP

Unit 1: Paper 1 Intensive Revision (Weeks 1-4)

WEEK 1: Data Representation & Transmission (Revision)

Topic coverage: Syllabus 1.1 -- 1.3 (binary, hexadecimal, units, ASCII/Unicode,

bitmap images, sound sampling, serial/parallel, error detection, encryption)

Lesson 1 (40 min): Binary & Hexadecimal Quick Review

- Starter (5 min): Convert these numbers race (binary to denary, 8 questions)
- Main (25 min): Binary addition, hexadecimal conversion, two's complement recap

>> Concept check: "Why does a computer use binary?" (2 marks)

>> Worked example: 8-bit two's complement representation of -73

>> Practice: 6 conversion questions (individual, mini-whiteboards)

- Plenary (10 min): Hexadecimal colour code activity (#FF5733 = ?)
- Homework: Syllabus 1 revision worksheet (15 questions)

Lesson 2 (40 min): Images, Sound, Compression

- Starter (5 min): "What affects file size?" mind map on whiteboard
- Main (25 min): Bitmap formula (width x height x colour depth), sampling rate,

sample resolution, lossy vs lossless compression

>> Worked example: Calculate file size of 800x600 image at 24-bit colour

>> Exam question walkthrough: 4-mark sound sampling question

>> Practice: Calculate storage requirements (3 scenarios)

- Plenary (10 min): Lossy or Lossless? -- card sort activity
- Homework: Past paper question (images) + self-mark using mark scheme

Lesson 3 (40 min): Data Transmission & Error Detection

- Starter (5 min): Serial vs Parallel -- quick comparison table

- Main (25 min): Simplex/duplex, parity checks (odd/even), checksums, echo checking, ARQ; distinction between error DETECTION and error CORRECTION

>> Exam technique: "State TWO methods of error detection" (2 marks)

>> Worked example: Calculate odd parity bit for 0110101

>> Practice: 4 short-answer questions

- Plenary (10 min): "Explain how parity checking works" -- paired explanation
- Homework: Error detection worksheet

Lesson 4 (40 min): Encryption & Timed Practice

- Starter (5 min): Plain text vs cipher text -- examples
- Main (25 min): Symmetric vs asymmetric encryption, private/public keys, SSL/TLS, digital signatures, malware types (virus, worm, trojan, spyware)

>> Key concept: "Why is asymmetric encryption more secure?" (2 marks)

>> Practice: 5 mixed questions from Syllabus 1

- Plenary (10 min): Timed practice: 10-mark mini-test (10 minutes)
- Homework: Complete any unfinished work + review mark scheme

Week 1 Assessment: End-of-week 15-mark test (Syllabus 1 only)

Resources: Syllabus 1 revision booklet, past paper questions (0478/11, 0478/12), mini-whiteboards, mark scheme booklet

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WEEK 2: Hardware & Software (Revision)

Topic coverage: Syllabus 1.4 -- 1.5 (CPU architecture, memory, storage, I/O, operating systems, utility software, translators)

Lesson 1 (40 min): CPU Architecture & Fetch-Decode-Execute

- Starter (5 min): Label the CPU diagram (registers, ALU, CU, buses)
- Main (25 min): Von Neumann architecture, registers (PC, ACC, MAR, MDR, CIR), fetch-decode-execute cycle step-by-step

>> Worked example: Trace through 3 cycles of FDE with sample instructions

>> Exam technique: "Describe the purpose of the MAR" (2 marks)

>> Practice: 4 questions on CPU components

- Plenary (10 min): FDE cycle -- order the steps card activity
- Homework: Draw and annotate the FDE cycle diagram

Lesson 2 (40 min): Memory & Storage

- Starter (5 min): RAM vs ROM -- think-pair-share
- Main (25 min): RAM (volatile, read/write), ROM (non-volatile, read-only), cache (faster, smaller, between CPU and RAM), virtual memory, secondary storage (HDD, SSD, optical, flash), data capacity units (bit to TB)

>> Worked example: Why is cache faster than RAM? (2 marks)

>> Comparison table: HDD vs SSD (speed, cost, capacity, durability)

>> Practice: "Explain why a computer needs both RAM and ROM" (3 marks)

- Plenary (10 min): Storage scenario -- "Which storage for a gaming PC?"
- Homework: Memory hierarchy diagram + explanations

Lesson 3 (40 min): Operating Systems & Utility Software

- Starter (5 min): Name 5 functions of an OS -- list on whiteboard
- Main (25 min): Memory management, multitasking, user interface, security,

disk defragmentation, backup software, compression, virus checking,

disk formatting; differences between freeware, shareware, open source

>> Exam question: "Explain the purpose of disk defragmentation" (3 marks)

>> Venn diagram: Freeware vs Shareware vs Open Source

>> Practice: OS functions matching activity

- Plenary (10 min): "Give two differences between freeware and shareware"
- Homework: Utility software research + 5 exam-style questions

Lesson 4 (40 min): Translators & Timed Practice

- Starter (5 min): Compiler vs Interpreter vs Assembler -- quick quiz
- Main (25 min): High-level vs low-level languages, compiler (produces

standalone executable, reports ALL errors at end), interpreter (executes

line by line, reports errors immediately), assembler (converts assembly

to machine code), stages of compilation (lexical analysis, syntax analysis,

code generation, optimisation)

>> Worked example: "Why is an interpreter better for program development?"

>> Exam technique: Stages of compilation -- explain each (4 marks)

>> Practice: 5 mixed questions from Syllabus 1.4-1.5

- Plenary (10 min): 10-mark timed mini-test
- Homework: Review all mistakes from timed test

Week 2 Assessment: End-of-week 15-mark test (Syllabus 1.4-1.5)

WEEK 3: The Internet & Security (Revision)

Topic coverage: Syllabus 2.1 -- 2.3 (networking, IP addressing, domains,

MAC addresses, client-server, web technologies, security, cookies)

Lesson 1 (40 min): Networks, IP Addressing & Packet Switching

- Starter (5 min): LAN vs WAN -- give an example of each
- Main (25 min): Star and bus topologies (diagrams, advantages, disadvantages),

IP addresses (IPv4 format, IPv6), MAC addresses, packet switching

(how data is split, routed, reassembled), routers, switches, NICs

>> Diagram: Draw a star network with 4 devices, label all components

>> Exam question: "Explain how packet switching works" (4 marks)

>> Practice: IP address validation questions

- Plenary (10 min): "What is the difference between a router and a switch?"
- Homework: Network topology comparison table

Lesson 2 (40 min): Internet Technologies

- Starter (5 min): URL anatomy -- label the parts
- Main (25 min): DNS (domain name system), domain name structure, web browsers,

web servers, FTP, HTTP/HTTPS, HTML, SMTP/POP3/IMAP (email protocols),

web vs internet distinction

>> Worked example: "Explain the role of a DNS server" (3 marks)

>> Exam technique: "Describe what happens when you type a URL" (4 marks)

>> Practice: Protocol matching (HTTP, FTP, SMTP, IMAP, HTTPS)

- Plenary (10 min): Client-server model diagram activity
- Homework: Internet technologies crossword + questions

Lesson 3 (40 min): Security & Cookies

- Starter (5 min): Security threat brainstorm (2 minutes, list as many as possible)
- Main (25 min): Phishing, pharming, hacking, malware, spyware, firewall,

antivirus, encryption (symmetric/asymmetric), authentication (biometrics,

passwords, two-factor), cookies (purpose, privacy concerns)

>> Case study: "A bank wants to protect customer data -- suggest 3 methods"

>> Exam question: "Describe what a cookie is and explain a concern" (3 marks)

>> Practice: Security scenarios -- suggest protections

- Plenary (10 min): "SSL/TLS" -- explain with diagram (2 marks)
- Homework: Security threats and protections matching worksheet

Lesson 4 (40 min): Timed Practice & Exam Technique

- Starter (5 min): Syllabus 2 quick-fire questions (10 questions, 30 seconds each)
- Main (25 min): Full Section A-style practice (15 marks, 20 minutes timed)
  - 5 short-answer questions on Syllabus 2 topics
  - Self-mark using provided mark scheme
  - Common errors class discussion
- Plenary (10 min): Exam technique tip: "Read the command word carefully"
  - State = 1-2 lines, no explanation
  - Describe = give details of how it works
  - Explain = give reasons WHY
- Homework: Review all Syllabus 2 mistakes

Week 3 Assessment: End-of-week 20-mark test (Syllabus 2 only)

WEEK 4: Automated Technologies & Paper 1 Mixed Practice

Topic coverage: Syllabus 3 (robotics, AI, expert systems, emerging tech)

Lesson 1 (40 min): Robotics & Automated Systems

- Starter (5 min): "Name 5 devices that contain a sensor" -- list
- Main (25 min): Sensors (temperature, pressure, light, motion, sound, pH), actuators, microprocessors, ADC/DAC, applications (washing machines, security systems, temperature control), benefits and drawbacks

  >> Exam question: "Explain the role of sensors and actuators" (4 marks)

  >> Scenario: "Design a greenhouse control system" -- labelled diagram

  >> Practice: Sensor selection (which sensor for which application?)

- Plenary (10 min): ADC vs DAC -- compare in table
- Homework: Automated systems worksheet

Lesson 2 (40 min): Artificial Intelligence & Expert Systems

- Starter (5 min): AI examples in daily life (Siri, recommendations, etc.)
- Main (25 min): AI definition, machine learning, expert systems (knowledge base, inference engine, rules base), applications (medical diagnosis, chess, voice recognition), benefits and limitations

  >> Exam question: "Describe the components of an expert system" (3 marks)

  >> Case study: "Should AI replace human doctors?" -- structured debate

  >> Practice: Expert system flow diagram activity

- Plenary (10 min): "Explain one benefit and one risk of AI" (4 marks)
- Homework: AI and expert systems exam questions

Lesson 3 (40 min): Emerging Technologies

- Starter (5 min): "What technology do you think will be important in 2030?"
- Main (25 min): Emerging technologies (autonomous vehicles, wearable tech, 3D printing, virtual/augmented reality, quantum computing, blockchain), digital currency, biometrics, social implications

  >> Discussion: Ethical considerations of emerging technology

  >> Exam technique: "Evaluate the impact of autonomous vehicles" (6 marks)

  >> Practice: Impact analysis grid (social, economic, ethical)

- Plenary (10 min): Key definitions quiz
- Homework: Emerging technologies research + questions

Lesson 4 (40 min): Paper 1 Full Timed Practice (30 marks)

- This lesson: Full 30-mark Paper 1 practice test (40 minutes)
- Questions drawn from Syllabus 1, 2, and 3
- Mixed format: short answer, structured, one extended question
- Self-mark using provided mark scheme in following lesson

Lesson 5 (40 min): Review & Self-Marking

- Hand back marked papers (or self-mark if not teacher-marked)
- Common errors discussion (15 min)
- Individual error log update (10 min)
- Target setting for next unit (10 min)
- Homework: Revise weakest 3 topics from Unit 1

Week 4 Assessment: 30-mark Paper 1 mixed practice test

Unit 1 Assessment Summary: 80 marks total across weekly tests + 30-mark practice

Unit 2: Paper 2 Intensive Revision (Weeks 5-8)

WEEK 5: Algorithms & Pseudocode (Revision)

Topic coverage: Syllabus 1.1 -- 1.2 (algorithm design, pseudocode, flowcharts)

Lesson 1 (40 min): Algorithm Design & Pseudocode Basics

- Starter (5 min): "What is an algorithm?" -- paired definition
- Main (25 min): Pseudocode syntax recap (DECLARE, SET, INPUT, OUTPUT, IF...

THEN...ELSE...ENDIF, REPEAT...UNTIL, WHILE...DO...ENDWHILE, FOR...

TO...NEXT, STEP), flowchart symbols, decomposing problems

>> Worked example: Write pseudocode to find the largest of 3 numbers

>> Practice: Convert flowchart to pseudocode (2 exercises)

>> Exam technique: "Write an algorithm to..." -- structuring your answer

- Plenary (10 min): Pseudocode syntax matching activity
- Homework: Write pseudocode for 3 simple problems

Lesson 2 (40 min): Selection & Iteration Structures

- Starter (5 min): IF vs CASE -- when to use each?
- Main (25 min): Nested IF statements, CASE statements, FOR loops (count-

controlled), WHILE loops (condition-controlled), REPEAT...UNTIL loops,

choosing the right loop structure, boundary values

>> Worked example: Pseudocode for number guessing game

>> Practice: Trace through a given loop (table provided)

>> Exam question: "Write an algorithm that validates user input" (4 marks)

- Plenary (10 min): Loop type selection -- which loop for which scenario?
- Homework: Loop practice worksheet (write + trace)

Lesson 3 (40 min): Arrays & String Handling

- Starter (5 min): "Why are arrays useful?" -- 2 minute discussion
- Main (25 min): 1D arrays (declaring, initialising, accessing elements),

2D arrays (rows and columns), array traversal (FOR loop), searching

(linear search algorithm), string operations (length, substring,

concatenation, ASCII conversion)

>> Worked example: Linear search pseudocode with trace table

>> Practice: "Write pseudocode to find the average of 10 numbers in an array"

>> Exam question: "Write pseudocode to count occurrences of 'A' in a string"

- Plenary (10 min): Array index quiz (what is the index of the 5th element?)
- Homework: Arrays and strings worksheet

Lesson 4 (40 min): File Handling & Timed Practice

- Starter (5 min): "Name 3 reasons to use files" -- brainstorm
- Main (25 min): OPENFILE, READFILE, WRITEFILE, CLOSEFILE, CSV format,

reading until EOF, writing data to files, practical applications

>> Worked example: Read names from file and count how many

>> Exam technique: File handling in pseudocode -- common syntax errors

>> Practice: 5 questions on file operations

- Plenary (10 min): 10-mark timed mini-test
- Homework: Review all mistakes

Week 5 Assessment: 20-mark algorithms test (write + trace)

WEEK 6: Programming Constructs & Subprograms (Revision)

Topic coverage: Syllabus 1.3 -- 1.5 (operators, functions, procedures, parameters)

Lesson 1 (40 min): Operators & Expressions

- Starter (5 min): Arithmetic operators quick-fire (+, -, *, /, MOD, DIV, ^)
- Main (25 min): Relational operators (=, <>, <, >, <=, >=), logical operators

(AND, OR, NOT), operator precedence, Boolean expressions, truth tables

for compound conditions

>> Worked example: Evaluate (x > 5) AND (x < 20) when x = 12

>> Practice: Evaluate 6 Boolean expressions

>> Exam question: "Write a condition to check if a number is 1-100 AND even"

- Plenary (10 min): Operator precedence puzzle
- Homework: Operators worksheet (20 questions)

Lesson 2 (40 min): Functions & Procedures

- Starter (5 min): Function vs Procedure -- what's the difference?
- Main (25 min): FUNCTION...ENDFUNCTION, PROCEDURE...ENDPROCEDURE, parameters

(by value), return values, local variables, global variables, scope,

advantages of modular design (reusability, readability, easier testing)

>> Worked example: Write a function to calculate the area of a circle

>> Practice: "Write a procedure that displays a menu" (no return value)

>> Exam question: "Explain two advantages of using subprograms" (4 marks)

- Plenary (10 min): Scope quiz -- local or global variable?
- Homework: Functions and procedures worksheet

Lesson 3 (40 min): Code Reading & Trace Tables

- Starter (5 min): "Trace this code" -- simple 3-line example
- Main (25 min): Reading pseudocode (identifying inputs, outputs, purpose),

completing trace tables (columns for variables, conditions, outputs),

identifying logic errors, dry running algorithms

>> Worked example: Complete trace table for a WHILE loop (provided)

>> Practice: 3 trace table exercises (increasing difficulty)

>> Exam technique: "Show your working" -- marks for partial trace tables

- Plenary (10 min): "What does this algorithm do?" -- code reading challenge
- Homework: Trace table practice (3 problems)

Lesson 4 (40 min): Code Writing Practice & Timed Test

- Starter (5 min): Planning grid activity (given problem, plan solution)
- Main (25 min): Structured code writing practice
  - Q1: Write a function to validate a password (4 marks, 6 min)
  - Q2: Write a procedure to display array contents (3 marks, 5 min)
  - Q3: Write an algorithm to find the smallest value in a 2D array (5 marks, 8 min)
  - Q4: Write a program to read a file and calculate average (5 marks, 8 min)
- Plenary (10 min): Peer assessment using mark scheme
- Homework: Review errors + practise weakest area

Week 6 Assessment: 20-mark code reading/writing test

WEEK 7: Databases, SQL & Boolean Logic (Revision)

Topic coverage: Syllabus 2.1 -- 2.2 (databases, SQL, Boolean logic)

Lesson 1 (40 min): Database Concepts

- Starter (5 min): "Why use a database instead of a spreadsheet?"
- Main (25 min): Flat-file databases (problems: redundancy, inconsistency),

relational databases (tables, relationships, primary keys, foreign keys),

entity-relationship diagrams (1:1, 1:M, M:N), normalisation (1NF, 2NF, 3NF),

advantages of relational databases

>> Exam question: "Explain two problems with flat-file databases" (4 marks)

>> Practice: Draw an ER diagram for a school database

>> Worked example: Identify primary and foreign keys in given tables

- Plenary (10 min): Database terminology matching
- Homework: ER diagram exercise

Lesson 2 (40 min): SQL Queries

- Starter (5 min): SQL keyword recap (SELECT, FROM, WHERE, ORDER BY)
- Main (25 min): SELECT statements, WHERE clauses (comparison operators, LIKE, BETWEEN, IN), ORDER BY (ASC, DESC), aggregate functions (COUNT, SUM, AVG, MAX, MIN), GROUP BY, JOIN (INNER JOIN), wildcard characters (% and _)

  >> Worked example: "Write SQL to find all students in Class 10A, sorted by name"

  >> Practice: 8 SQL questions on provided database schema

  >> Exam question: "Write SQL to count how many orders were placed in June"
- Plenary (10 min): SQL error spotting (find mistakes in given queries)
- Homework: SQL practice worksheet (10 questions)

Lesson 3 (40 min): Boolean Logic & Logic Gates

- Starter (5 min): AND, OR, NOT -- real-life examples (not computing)
- Main (25 min): Logic gates (AND, OR, NOT, NAND, NOR, XOR), truth tables for all gates, combining logic gates (drawing circuits, writing expressions), logic diagram to Boolean expression, Boolean expression to logic diagram, applications (security systems, CPU control, washing machines)

  >> Worked example: Draw a circuit for A AND (NOT B) -- truth table

  >> Practice: Complete truth tables for 3 combined circuits

  >> Exam question: "Draw a logic circuit for (A OR B) AND C" (3 marks)
- Plenary (10 min): Logic gate symbol matching
- Homework: Boolean logic worksheet (truth tables + circuits)

Lesson 4 (40 min): Boolean Applications & Simplification

- Starter (5 min): "Where in a computer is Boolean logic used?"
- Main (25 min): Boolean algebra (simplification rules: A AND 1 = A, A OR 0 = A, A AND A = A, A OR A = A, A AND NOT A = 0, A OR NOT A = 1, NOT(NOT A) = A, De Morgan's laws), deriving Boolean expressions from truth tables, applying logic to search engines and programming

  >> Worked example: Simplify (A AND B) OR (A AND B) -- answer = A AND B

  >> Practice: Simplify 4 Boolean expressions

  >> Exam question: "Write the Boolean expression for this truth table" (4 marks)
- Plenary (10 min): Boolean algebra rules quiz
- Homework: Boolean simplification worksheet

Week 7 Assessment: 20-mark databases/SQL/Boolean test

WEEK 8: Scenario Questions & Paper 2 Mixed Practice

Lesson 1 (40 min): Understanding Scenario Questions

- Starter (5 min): "What makes a good answer to a 15-mark question?"
- Main (25 min): Structure of the 15-mark scenario question, reading the

scenario carefully, identifying requirements, breaking down the problem, planning the answer (pseudocode + explanation), mark allocation analysis (typically: design 3 marks, code 8 marks, testing 2 marks, comments 2 marks)

>> Worked example: Walk through a sample 15-mark question together

>> Practice: Plan (only) a solution to a scenario question (no coding yet)

>> Exam technique: "Read the scenario THREE times before writing"

- Plenary (10 min): Mark scheme analysis -- how are marks awarded?
- Homework: Read and plan 2 scenario questions from past papers

Lesson 2 (40 min): Writing Scenario Answers

- Starter (5 min): Review planning homework -- share best plans
- Main (25 min): Writing structured pseudocode for scenario questions,

including meaningful variable names, comments, input validation, modular design (using functions/procedures), testing strategy

>> Worked example: Write full answer to 8-mark scenario question

>> Practice: Write answer to 12-mark scenario question (timed, 20 min)

>> Peer assessment using mark scheme

- Plenary (10 min): Common errors in scenario answers
- Homework: Complete any unfinished scenario answers

Lesson 3 (40 min): Paper 2 Mixed Practice (Part 1)

- Full 25-mark Section A + B practice (35 minutes timed)
- Short answer algorithms questions
- Code reading and trace table completion
- Code writing exercises
- Self-mark using mark scheme

Lesson 4 (40 min): Paper 2 Mixed Practice (Part 2)

- 15-mark scenario question practice (25 minutes timed)
- Self-mark using mark scheme
- Individual gap analysis: Which Paper 2 topics need more work?
- Target setting for Mock Series 1

Week 8 Assessment: 40-mark Paper 2 practice test

Unit 2 Assessment Summary: 60 marks across weekly tests + 40-mark practice

Unit 3: Mock Series 1 (Weeks 9-10)

WEEK 9: MOCK EXAM 2 -- PAPER 1

Duration: 1 hour 45 minutes (105 minutes)

Marks: 75 marks

Format: Full Paper 1 (Theory)

Monday: Exam conditions mock examination

- Full exam setup (individual desks, no notes, no phones)
- Paper covers ALL Syllabus Topics 1-6
- Section A: Short answer (approximately 30 marks)
- Section B: Structured questions (approximately 30 marks)
- Section C: Extended answer question (approximately 15 marks)

Wednesday: Teacher marking / peer marking preparation

Friday: Paper returned + review

- Class-level analysis: Which questions were hardest?
- Individual analysis: Each student updates their progress tracker
- Target setting: 3 specific targets based on errors

Mock 2 Paper 1: See Assessment Pack (03_VPA_CS_Y10S2_Assessment_Pack.txt)

for full question paper and mark scheme.

WEEK 10: MOCK EXAM 3 -- PAPER 2

Duration: 1 hour 45 minutes (105 minutes)

Marks: 75 marks

Format: Full Paper 2 (Problem Solving & Programming)

Monday: Exam conditions mock examination

- Full exam setup
- Paper covers ALL Syllabus Topics 7-10
- Section A: Short answer (approximately 25 marks)
- Section B: Code reading and writing (approximately 35 marks)
- Section C: 15-mark scenario question

Wednesday: Teacher marking

Friday: Paper returned + review

- Class-level analysis
- Individual progress tracker update
- Target setting for targeted revision unit

Mock 3 Paper 2: See Assessment Pack for full question paper and mark scheme.

Unit 4: Gap Analysis & Targeted Revision (Weeks 11-14)

WEEK 11: Gap Analysis & Individual Target Setting

Lesson 1 (40 min): Mock 2 & 3 Results Analysis

- Starter (5 min): "What do your results tell you?" -- reflection
- Main (25 min): Individual gap analysis using templates

>> Compare Mock 1 vs Mock 2 (Paper 1) -- where did you improve/decline?

>> Compare previous Paper 2 performance vs Mock 3

>> Identify TOP 3 weak topics for each student

>> Class-level analysis: Which topics did MOST students struggle with?

- Plenary (10 min): Share common gaps -- class revision priority list
- Homework: Complete personal gap analysis template

Lesson 2 (40 min): Target Setting & Revision Planning

- Starter (5 min): "SMART targets" -- what makes a good target?
- Main (25 min): Writing 3 SMART revision targets per student

>> Target 1: Topic to improve (e.g., "Improve SQL queries from 2/8 to 6/8")

>> Target 2: Skill to develop (e.g., "Complete trace tables accurately")

>> Target 3: Exam technique (e.g., "Read all scenario info before coding")

>> Each student creates a revision timetable for Weeks 11-14

- Plenary (10 min): Partner check -- is your partner's target SMART?
- Homework: Begin revision on Target 1 topic

Lesson 3-4 (80 min): Grouped Revision Session 1

- Students divided into groups by weak area:

Group A: Data Representation & Binary

Group B: Programming & Pseudocode

Group C: Databases & SQL

Group D: Boolean Logic & Trace Tables

- Rotation activities (2 x 40 min sessions)
- Each group: teacher-led revision + practice questions + self-check

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WEEK 12: Targeted Revision -- Paper 1 Weak Areas

Lesson 1-2 (80 min): Hardware & Software Deep Dive

- For students weak in Syllabus 1.4-1.5
- FDE cycle step-by-step practice
- Memory hierarchy card sort
- Compiler stages sequencing activity
- Past paper questions focus

Lesson 3-4 (80 min): Internet & Security Deep Dive

- For students weak in Syllabus 2
- Network diagram drawing practice
- Protocol matching games
- Security scenario questions
- Exam technique: "Explain" vs "Describe" vs "State"

WEEK 13: Targeted Revision -- Paper 2 Weak Areas

Lesson 1-2 (80 min): Algorithms & Programming Deep Dive

- For students weak in Syllabus 1 (Paper 2 aspects)
- Pseudocode writing drills
- Trace table competitions
- Code reading challenges
- Functions and procedures practice

Lesson 3-4 (80 min): Databases & Boolean Deep Dive

- SQL query writing practice (10 questions)
- Boolean algebra simplification
- Logic circuit drawing
- Mixed practice exam

WEEK 14: Mixed Practice & Confidence Building

Lesson 1 (40 min): Paper 1 Speed Practice

- 20 marks in 25 minutes (faster than exam pace)
- Build speed and automaticity
- Self-mark + error log

Lesson 2 (40 min): Paper 2 Speed Practice

- 20 marks in 25 minutes
- Focus: quick code reading and writing
- Self-mark + error log

Lesson 3 (40 min): Mixed Topic Relay

- Team competition: 5 stations, 5 topics, 5 minutes each
- Scoring system + prizes
- High-energy revision activity

Lesson 4 (40 min): Confidence Check & Mock Prep

- Individual confidence rating (1-5) for all topics
- Last-minute revision priorities identified
- Mock 4 preparation: exam technique reminder
- Relaxation techniques for exam anxiety

Week 14 Assessment: 20-mark mixed speed test

Unit 5: Mock Series 2 (Weeks 15-16)

WEEK 15: MOCK EXAM 4 -- PAPER 1

Duration: 1 hour 45 minutes (105 minutes)

Marks: 75 marks

Format: Full Paper 1 (Theory) -- DIFFERENT from Mocks 1 and 2

Monday: Exam conditions mock examination

- Strict exam conditions (as per real IGCSE)
- Fresh paper covering all Syllabus Topics 1-6
- Designed to be at or slightly above expected exam difficulty

Wednesday: Teacher marking

Friday: Paper returned + detailed review

- Mock 1 vs Mock 2 vs Mock 4 progression analysis
- Celebrate improvements
- Identify any remaining gaps
- Final target setting for last 2 weeks

Mock 4 Paper 1: See Assessment Pack for full question paper and mark scheme.

WEEK 16: MOCK EXAM 5 -- PAPER 2

Duration: 1 hour 45 minutes (105 minutes)

Marks: 75 marks

Format: Full Paper 2 (Problem Solving & Programming) -- DIFFERENT from Mocks 1 and 3

Monday: Exam conditions mock examination

- Strict exam conditions
- Fresh paper covering all Syllabus Topics 7-10
- Includes 15-mark scenario question

Wednesday: Teacher marking

Friday: Paper returned + final analysis

- Mock 1 vs Mock 3 vs Mock 5 progression analysis
- Overall Paper 2 progress review
- Individual "exam readiness" rating
- Transition to Unit 6: Final Preparation

Mock 5 Paper 2: See Assessment Pack for full question paper and mark scheme.

Unit 6: Final Preparation (Weeks 17-18)

WEEK 17: Revision Clinic & Exam Technique Polish

Lesson 1 (40 min): Last-Minute Paper 1 Priorities

- Starter (5 min): "The 10 most important facts for Paper 1" -- quiz
- Main (25 min): High-yield topics (most marks, most common):

>> Fetch-decode-execute cycle (guaranteed question)

>> Network topology and protocols (frequent)

>> Security methods and threats (always examined)

>> Data representation calculations (binary/hex/file sizes)

>> Emerging technology evaluation questions

- Plenary (10 min): Quick-fire definitions (20 terms, rapid response)
- Homework: Revise top 5 weakest topics

Lesson 2 (40 min): Last-Minute Paper 2 Priorities

- Starter (5 min): Pseudocode syntax speed test
- Main (25 min): High-yield programming topics:

>> Trace tables (common, easy marks if practised)

>> Pseudocode writing (functions, loops, arrays)

>> SQL queries (SELECT, WHERE, ORDER BY, aggregate functions)

>> Boolean algebra (simplification, truth tables)

>> 15-mark scenario question strategy

- Plenary (10 min): "One thing I must remember about..." each student shares
- Homework: Write one perfect function and one perfect procedure

Lesson 3 (40 min): Exam Technique Masterclass

- Starter (5 min): "Common mistakes that cost marks" -- list
- Main (25 min): Exam technique reminders:

>> Time management: 75 marks / 105 minutes = 1.4 min per mark

>> Read the COMMAND WORD: State (1 line), Describe (how it works), Explain (reasons why), Evaluate (advantages AND disadvantages)

>> Show ALL working for calculation questions

>> For coding: add comments, use meaningful variable names

>> Check your answers: 5 minutes at the end for review

>> If stuck: mark the question, move on, come back later

- Plenary (10 min): "Command word" matching game
- Homework: Review all past mock errors one final time

Lesson 4 (40 min): Individual Revision Surgery

- Open revision session
- Students work on their own weak areas
- Teacher circulates for 1-to-1 support
- Peer tutoring: strong students help weaker students
- Resource stations: past papers, mark schemes, revision guides, flashcards

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WEEK 18: Confidence Building & Exam Day Preparation

Lesson 1 (40 min): "You Know More Than You Think"

- Positive affirmation activity
- Review of how far students have come (Y9 to now)
- Success stories from previous cohorts

- Individual "I can..." statement writing
- Create personal "cheat sheet" (one A4 page of key facts)

Lesson 2 (40 min): Exam Day What-to-Bring & Logistics

- Exam day checklist (see Assessment Pack)
- What to expect in the exam room
- Timing and schedule
- What to do if: you feel panicked / you don't understand a question / you run out of time / you make a mistake
- Q&A session

Lesson 3 (40 min): Final Paper 1 Warm-Up

- 20-mark Paper 1 practice (25 minutes)
- Immediate feedback
- Last confidence boost
- Reminder: "Trust your revision"

Lesson 4 (40 min): Final Paper 2 Warm-Up

- 20-mark Paper 2 practice (25 minutes)
- Immediate feedback
- Final "good luck" and celebration of hard work
- Reminder of exam dates and requirements

END OF COURSE: Students enter IGCSE examination period confident, well-prepared, and ready to demonstrate their knowledge.

SECTION 4: ASSESSMENT CALENDAR

SEMESTER ASSESSMENT OVERVIEW

Week	Assessment	Type	Marks	Duration
W1	End-of-week test (Syllabus 1)	Written	15	20 min
W2	End-of-week test (Syllabus 1.4-1.5)	Written	15	20 min
W3	End-of-week test (Syllabus 2)	Written	20	25 min
W4	Paper 1 mixed practice test	Written	30	35 min
W5	Algorithms test	Written	20	25 min
W6	Code reading/writing test	Written	20	25 min
W7	Databases/SQL/Boolean test	Written	20	25 min
W8	Paper 2 mixed practice test	Written	40	50 min
W9	MOCK EXAM 2 -- PAPER 1	Formal Mock	75	1h 45m
W10	MOCK EXAM 3 -- PAPER 2	Formal Mock	75	1h 45m
W11	Gap analysis review	Analysis	--	--
W14	Mixed speed test	Written	20	25 min
W15	MOCK EXAM 4 -- PAPER 1	Formal Mock	75	1h 45m
W16	MOCK EXAM 5 -- PAPER 2	Formal Mock	75	1h 45m

W17	Revision clinic	Ongoing	--	--
W18	Final warm-up tests	Written	40	50 min

MOCK EXAM TRACKING TEMPLATE (MOCK 1 THROUGH MOCK 5)

STUDENT NAME: ________________________________ CLASS: ____________

PAPER 1 TRACKING (Theory -- 75 marks)

Paper	Date	Raw Mark	/75	Percentage	Grade	Notes
Mock 1	Y10S1 W17	_________	75	_______%	___	____________
Mock 2	Y10S2 W9	_________	75	_______%	___	____________
Mock 4	Y10S2 W15	_________	75	_______%	___	____________

PROGRESS MOCK 1 -> MOCK 2: ______ marks change (______% change)

PROGRESS MOCK 2 -> MOCK 4: ______ marks change (______% change)

OVERALL PROGRESS M1 -> M4: ______ marks change (______% change)

PAPER 2 TRACKING (Problem Solving -- 75 marks)

Paper	Date	Raw Mark	/75	Percentage	Grade	Notes
Mock 1	Y10S1 W18	_________	75	_______%	___	____________
Mock 3	Y10S2 W10	_________	75	_______%	___	____________
Mock 5	Y10S2 W16	_________	75	_______%	___	____________

PROGRESS MOCK 1 -> MOCK 3: ______ marks change (______% change)

PROGRESS MOCK 3 -> MOCK 5: ______ marks change (______% change)

OVERALL PROGRESS M1 -> M5: ______ marks change (______% change)

TOTAL SCORE TRACKING (Paper 1 + Paper 2 = 150 marks)

Paper	Paper 1	Paper 2	TOTAL	/150	%	Overall Grade
Mock 1	___/75	___/75	___/150	150	___%	___
Mock 2	___/75	--	___/75	75	___%	___
Mock 3	--	___/75	___/75	75	___%	___
Mock 4	___/75	--	___/75	75	___%	___
Mock 5	--	___/75	___/75	75	___%	___

GRADE BOUNDARIES (Typical IGCSE 0478 -- for reference)

A*: 80-100%   A: 70-79%   B: 60-69%   C: 50-59%

D: 40-49%   E: 30-39%   F: 20-29%   G: 10-19%

(Note: Actual grade boundaries set by CIE after each exam series)

TARGET GRADE: ____________ PREDICTED GRADE: ____________

TEACHER SIGNATURE: ____________________________ DATE: ____________

SECTION 5: STUDENT PROFILE

STUDENT DEMOGRAPHICS

School: Victoria Park Academy, Shenzhen, China

Programme: CIE IGCSE Computer Science (0478)

Year Group: Year 10 (Final semester before IGCSE exams)

Language Profile: EAL (English as Additional Language)

Ability Range: Mixed ability (predicted grades A* to G)

Class Size: Typically 15-25 students

Lesson Duration: 40 minutes per period

Lessons per Week: Typically 4-5 periods

IMPLICATIONS FOR TEACHING

EAL CONSIDERATIONS:

- All key terms provided in English and Chinese
- Sentence starters provided for extended answers
- Visual aids and diagrams used wherever possible
- Key vocabulary pre-taught before each revision topic
- Word walls displayed in classroom
- Bilingual glossary provided (50+ terms)
- Extra reading time allowed in practice (not mocks)
- Clear, simple instructions for all activities

MIXED ABILITY CONSIDERATIONS:

- Differentiated worksheets (Bronze/Silver/Gold)
- Extension questions for high-achieving students
- Scaffolded support for students working below target
- Peer tutoring and group work opportunities
- Individual revision plans based on gap analysis
- Flexible grouping for targeted revision sessions

40-MINUTE LESSON CONSIDERATIONS:

- Tight lesson structure (starter-main-plenary)
- Clear time signals for each activity
- No activity longer than 25 minutes
- Quick transitions between tasks
- Mini-plenaries to check understanding
- Efficient distribution/collection of resources

SECTION 6: STANDARD LESSON STRUCTURE

Every lesson follows this structure:

COMPONENT	TIME	PURPOSE
Starter	5 min	Activate prior knowledge, engage attention, link to previous learning, quick assessment
Main Activity	25 min	Core learning/revision content (teacher input, guided practice, independent work)
Plenary	10 min	Consolidate learning, check understanding,

preview next lesson, set homework

STARTER ACTIVITIES BANK (复习导入活动库)

- Quick quiz (5 questions on previous lesson)
- "Write 3 things you remember about..."
- True/False statements
- Vocabulary matching (English/Chinese)
- "Spot the error" in given code or definitions
- Mini-whiteboard race
- Think-Pair-Share
- "One question you'd like to ask about..."
- Timed recall challenge
- Connect the concept (how does X relate to Y?)

MAIN ACTIVITY FORMATS (主要活动形式)

- Teacher-led worked example (I do, we do, you do)
- Paired practice (solve together)
- Individual worksheet practice
- Past paper question walkthrough
- Mini-whiteboard whole-class response
- Group problem-solving challenge
- Code tracing on paper
- Diagram drawing and labelling
- Card sort (matching, sequencing, categorising)
- Peer teaching (explain to your partner)

PLENARY ACTIVITIES BANK (总结活动库)

- "3 things I learned, 2 things I found hard, 1 question I still have"
- Exit ticket (answer one question to leave)
- Self-assessment traffic lights (R/A/G)
- "Teach the person next to you..."
- Quick-fire questions around the room
- Predict the exam question
- Confidence rating (1-5) on today's topic
- Complete the sentence: "Today I learned that..."

SECTION 7: EAL SUPPORT CHECKLIST

FOR EVERY LESSON:

[ ] Key vocabulary displayed on board (English + Chinese)

[ ] Sentence starters provided for speaking/writing tasks

[ ] Visual aids/diagrams used to support verbal explanations

[ ] Instructions given verbally AND written on board

[ ] Key terms reviewed at start of lesson

[ ] One student asked to repeat instructions in their own words

FOR EVERY UNIT:

[ ] Bilingual vocabulary list provided at start of unit

[ ] Word wall updated with new terms

[ ] Chinese-speaking students seated near bilingual peers where helpful

[ ] Past paper questions with vocabulary pre-taught

[ ] Mark scheme language analysed and explained

FOR EVERY MOCK EXAM:

[ ] Exam vocabulary reviewed before the mock

[ ] Command words explained (State, Describe, Explain, Evaluate)

[ ] Rubric read through as a class before starting

[ ] Extra time considered for EAL learners (if school policy allows)

[ ] Post-exam: vocabulary from difficult questions reviewed

FOR INDIVIDUAL SUPPORT:

[ ] Bilingual dictionary available

[ ] Word bank for extended writing tasks

[ ] Sentence frames for "Explain" and "Evaluate" questions

[ ] Simplified language versions of complex questions (where appropriate)

[ ] One-to-one check-in for students struggling with language barriers

SECTION 8: DIFFERENTIATION FRAMEWORK

THREE-TIER SYSTEM (三层差异化体系)

BRONZE TIER (Core Support -- Target Grade C-E)

- Scaffolded worksheets with sentence starters
- Cloze (fill-in-the-blank) activities
- Matching and labelling exercises
- Multiple-choice practice questions
- Key facts lists provided
- Word banks for all writing tasks
- Teacher check-ins every 10 minutes
- Paired with supportive peer
- Focus: securing basic knowledge and understanding

SILVER TIER (Standard -- Target Grade B-A)

- Standard revision worksheets
- Short answer and structured questions
- Some extended writing with guidance
- Independent practice with mark scheme self-check
- Occasional teacher support
- Focus: developing exam technique and accuracy

GOLD TIER (Extension -- Target Grade A-A*)

- Challenge questions requiring application
- Full past paper questions under timed conditions

- Peer teaching responsibilities
- Research and extension tasks
- Evaluation and "design" questions
- Support other students (peer tutoring)
- Focus: perfecting exam technique, speed, and accuracy

DIFFERENTIATION BY TASK (任务差异化)

Same topic, different complexity:

Example - Binary addition:

Bronze: Add two 4-bit binary numbers

Silver: Add two 8-bit binary numbers, show carry

Gold: Add two 8-bit numbers, detect overflow, explain limitations

DIFFERENTIATION BY OUTCOME (成果差异化)

Same task, different expected outcomes:

Example - "Explain how the fetch-decode-execute cycle works"

Bronze: List the 3 stages with brief description (2 marks)

Silver: Describe each stage with register names (4 marks)

Gold: Explain with full register transfers and timing (6 marks)

DIFFERENTIATION BY SUPPORT (支持差异化)

Same task, different support levels:

- Max support: Word bank, sentence starters, worked example visible
- Medium support: Word bank only
- Min support: No scaffolding, independent work

SECTION 9: RESOURCE REQUIREMENTS

ESSENTIAL RESOURCES (必需资源)

PAST PAPERS:

- CIE 0478 Paper 1 (Theory): 2016-2024 series (all variants)
- CIE 0478 Paper 2 (Problem Solving): 2016-2024 series (all variants)
- Organised by topic for targeted revision
- Minimum 8 full Paper 1 papers and 8 full Paper 2 papers

MARK SCHEMES:

- Complete mark schemes for all past papers used
- Summary of common acceptable answers
- Student-friendly mark scheme versions (simplified language)

MINI-WHITEBOARDS:

- One per student
- Whiteboard pens (black, multiple colours)
- Erasers or tissues
- Used for: quick quizzes, whole-class checks, plenaries, races

INDIVIDUAL REVISION FOLDERS:

- One folder per student (kept in classroom)
- Divided into sections:

Section 1: Past paper questions by topic

Section 2: Mark schemes and model answers

Section 3: Error log (record of all mistakes)

Section 4: Progress tracker and target sheets

Section 5: Revision notes and mind maps

Section 6: Mock exam papers and results

PROGRESS TRACKERS:

- Wall display: class progress chart (anonymous or by student number)
- Individual: personal progress tracker in revision folder
- Digital: spreadsheet for teacher tracking

PRINTED MATERIALS (印刷材料)

- Revision booklets for each unit (teacher-created)
- Bilingual glossary (one per student)
- Exam technique guide (one per student)
- Formula sheet (binary/hex conversions, file size calculations)
- Pseudocode reference card (one per student)
- SQL syntax reference card
- Logic gate reference sheet
- Command word definitions poster
- "What to do if stuck" emergency guide

DIGITAL RESOURCES (数字资源)

- Interactive quizzes (Kahoot, Quizizz, Blooket)
- Binary/hex conversion practice websites
- SQL practice environments
- Past paper database (searchable by topic)
- Revision videos (recommended YouTube channels)
- Online flashcard sets (Quizlet)

PHYSICAL RESOURCES (实物资源)

- Coloured paper for revision activities
- Scissors, glue sticks (for card sorts)
- Highlighters (multiple colours per student)
- A4 paper for mock exams
- Stopwatch/timer for timed practice
- Sticky notes for exit tickets and feedback
- Laminator (for reusable card sorts)

SECTION 10: KEY BILINGUAL GLOSSARY (50+ TERMS)

EXAM VOCABULARY (考试词汇)

Multiple choice	选择题	A question with several possible answers

Structured question 结构化问题 A question with several parts building on each other

Extended answer 拓展回答 A longer written response (usually 4-6 marks)

Mark scheme 评分标准 The guide used to award marks for each question

Grade boundary 等级分数线 The minimum mark needed for each grade

Raw mark 原始分数 The actual marks scored

Percentage 百分比 Marks expressed as a proportion of 100

Trace table 跟踪表 A table showing how variable values change

Pseudocode 伪代码 An algorithm written in structured English

Flowchart 流程图 A diagram showing an algorithm using symbols

Scenario question 情景题 A long question based on a real-world situation

Command word 指令词 The word telling you what to do (State, Describe, Explain)

Timed practice 限时练习 Practice done under exam time limits

Past paper 历年真题 An exam paper from a previous year

Exam technique 考试技巧 Strategies for answering exam questions well

REVISION TERMS (复习词汇)

Gap analysis 差距分析 Finding which topics you don't know well

Target 目标 Something specific you want to improve

Consolidation 巩固 Strengthening what you already know

Mock exam 模拟考试 A practice exam under real exam conditions

Revision 复习 Going over work to prepare for an exam

Exam preparation 考试准备 Getting ready for the real exam

Time management 时间管理 Using exam time wisely

Weak area 薄弱领域 A topic or skill you find difficult

Strength 优势 A topic or skill you are good at

Progress tracker 进度跟踪表 A record showing improvement over time

Error log 错误日志 A list of mistakes to learn from

Self-assessment 自我评估 Judging your own work against criteria

Peer assessment 同伴评估 Judging a classmate's work

Marking scheme 评分方案 The correct answers and mark allocation

Model answer 模范答案 An excellent example answer

SYLLABUS 1: DATA REPRESENTATION (数据表示)

Binary 二进制 Number system using 0 and 1 (base-2)

Denary 十进制 Number system using 0-9 (base-10, decimal)

Hexadecimal 十六进制 Number system using 0-9 and A-F (base-16)

Two's complement 补码 Method for representing negative binary numbers

Bit 比特 Smallest unit of data (0 or 1)

Byte 字节 8 bits

Kilobyte (KB) 千字节 1024 bytes

Megabyte (MB) 兆字节 1024 kilobytes

Gigabyte (GB) 吉字节 1024 megabytes

Terabyte (TB) 太字节 1024 gigabytes

Pixel 像素 A single point in a digital image

Colour depth 颜色深度 Number of bits used to represent each pixel

Resolution 分辨率 Number of pixels in an image (width x height)

Sample rate 采样率 Number of samples taken per second (Hz)

Sample resolution 采样分辨率 Number of bits per sample

Metadata 元数据 Data about data (e.g., file type, date created)

Lossy compression 有损压缩 Reduces file size by removing some data permanently

Lossless compression 无损压缩 Reduces file size without losing any data

Parity check 奇偶校验 Error detection using an extra bit

Checksum 校验和 Error detection using a calculated value

Encryption 加密 Converting data to a secure code

Plain text 明文 Unencrypted, readable data

Cipher text 密文 Encrypted data

Symmetric encryption 对称加密 Same key for encryption and decryption

Asymmetric encryption 非对称加密 Different keys (public and private)

SYLLABUS 1: HARDWARE AND SOFTWARE (硬件与软件)

CPU 中央处理器 Central Processing Unit -- the "brain"

ALU 算术逻辑单元 Arithmetic Logic Unit -- performs calculations

CU 控制单元 Control Unit -- manages instruction execution

Register 寄存器 Small, fast memory locations in the CPU

Program Counter (PC) 程序计数器 Holds address of next instruction

Memory Address Reg (MAR) 内存地址寄存器 Holds address of data being accessed

Memory Data Reg (MDR) 内存数据寄存器 Holds data being transferred

Accumulator (ACC) 累加器 Stores results of calculations

Current Instr Reg (CIR) 当前指令寄存器 Holds instruction being decoded

Fetch-decode-execute 取指-译码-执行 The cycle the CPU follows to run programs

RAM	随机存取存储器	Random Access Memory -- volatile, read/write
ROM	只读存储器	Read Only Memory -- non-volatile, permanent
Cache	高速缓存	Fast memory between CPU and RAM
Virtual memory	虚拟内存	Using disk space as extra RAM
Secondary storage	辅助存储器	Long-term storage (HDD, SSD, etc.)
Hard disk drive (HDD)	硬盘驱动器	Magnetic storage with moving parts
Solid state drive (SSD)	固态硬盘	Flash memory storage, no moving parts
Optical storage	光存储	CDs, DVDs, Blu-ray discs
Flash memory	闪存	USB drives, memory cards
Operating system	操作系统	Software that manages the computer
Utility software	工具软件	Programs that maintain the computer
Compiler	编译器	Translates entire program to machine code
Interpreter	解释器	Translates and executes line by line
Assembler	汇编器	Translates assembly language to machine code
Source code	源代码	Program written in high-level language
Object code	目标代码	Machine code output from compilation
Freeware	免费软件	Software available at no cost
Shareware	共享软件	Try before you buy software
Open source	开源	Source code available for anyone to modify

SYLLABUS 2: INTERNET AND SECURITY (互联网与安全)

LAN	局域网	Local Area Network -- small geographical area
WAN	广域网	Wide Area Network -- spans large distances
Star topology	星型拓扑	All devices connected to central switch/hub
Bus topology	总线拓扑	All devices connected to single cable
Router	路由器	Connects networks and routes packets
Switch	交换机	Connects devices within a LAN
NIC	网络接口卡	Network Interface Card
IP address	IP 地址	Unique identifier for a device on a network
MAC address	MAC 地址	Unique hardware identifier of a NIC
Packet switching	分组交换	Data split into packets for transmission
Protocol	协议	Set of rules for data communication
DNS	域名系统	Domain Name System -- translates URLs to IPs
URL	统一资源定位符	Web address
HTTP	超文本传输协议	Protocol for web page transfer
HTTPS	安全超文本传输协议	Encrypted HTTP

SSL/TLS	安全传输层协议	Encryption for secure web communication
FTP	文件传输协议	Protocol for file transfer
SMTP	简单邮件传输协议	Sending email protocol
POP3	邮局协议	Receiving email protocol
IMAP	互联网消息访问协议	Advanced email receiving protocol
HTML	超文本标记语言	Language for creating web pages
Phishing	钓鱼攻击	Fake emails/websites to steal information
Pharming	网址嫁接	Redirecting to fake websites
Hacking	黑客攻击	Unauthorised access to computer systems
Firewall	防火墙	Hardware/software that blocks unauthorised access
Antivirus	杀毒软件	Detects and removes malware
Authentication	身份验证	Confirming identity (password, biometrics)
Cookie	Cookie	Small file stored by browser
Malware	恶意软件	Malicious software (virus, worm, trojan)

SYLLABUS 3: AUTOMATED TECHNOLOGIES (自动化技术)

Sensor	传感器	Device that detects physical conditions
Actuator	执行器	Device that produces physical movement
ADC	模数转换器	Analogue to Digital Converter
DAC	数模转换器	Digital to Analogue Converter
Microprocessor	微处理器	Single-chip CPU
Robotics	机器人技术	Design and use of robots
Artificial intelligence	人工智能	Computer systems that mimic human intelligence
Machine learning	机器学习	Systems that improve through experience
Expert system	专家系统	Computer system that mimics human expertise
Knowledge base	知识库	Store of facts and rules in an expert system
Inference engine	推理引擎	Part that applies rules to the knowledge base
Rules base	规则库	Collection of IF-THEN rules

SYLLABUS 1 (PAPER 2): ALGORITHMS AND PROGRAMMING (算法与编程)

Algorithm	算法	Step-by-step solution to a problem
Decomposition	分解	Breaking a problem into smaller parts
Abstraction	抽象	Removing unnecessary detail
Variable	变量	Named storage location for data
Constant	常量	Value that does not change
Array	数组	Collection of data items with same name
1D array	一维数组	Single list of values

2D array	二维数组	Table of values (rows and columns)
Selection	选择	Making a decision (IF, CASE)
Iteration	迭代	Repeating instructions (loop)
FOR loop	FOR 循环	Count-controlled iteration
WHILE loop	WHILE 循环	Condition-controlled iteration (check first)
REPEAT UNTIL	直到循环	Condition-controlled iteration (check last)
Function	函数	Subprogram that returns a value
Procedure	过程	Subprogram that performs a task
Parameter	参数	Value passed to a subprogram
Return value	返回值	Value sent back from a function
Local variable	局部变量	Only accessible within its subprogram
Global variable	全局变量	Accessible from anywhere in the program
Library routine	库例程	Pre-written code for common tasks
Random number	随机数	Unpredictable number for games/simulations
Rounding	四舍五入	Reducing decimal places
Truncation	截断	Removing decimal places without rounding
String handling	字符串处理	Operations on text (length, substring, etc.)
File handling	文件处理	Reading from and writing to files
CSV	逗号分隔值	Comma Separated Values file format
EOF	文件结尾	End Of File marker

SYLLABUS 2 (PAPER 2): DATABASES AND BOOLEAN LOGIC (数据库与布尔逻辑)

Database	数据库	Organised collection of data
Flat-file database	平面文件数据库	Single table database
Relational database	关系型数据库	Multiple linked tables
Table	表	Collection of related records
Record	记录	One row in a table
Field	字段	One column in a table
Primary key	主键	Unique identifier for each record
Foreign key	外键	Field that links to primary key in another table
Entity	实体	Object or concept stored in a database
Relationship	关系	Connection between entities
ER diagram	实体关系图	Visual representation of database structure
Normalisation	规范化	Organising data to reduce redundancy
SQL	结构化查询语言	Language for managing databases
SELECT	查询语句	Retrieve data from database

WHERE	条件子句	Filter records
ORDER BY	排序子句	Sort results
Wildcard	通配符	Special character for pattern matching
Aggregate function	聚合函数	COUNT, SUM, AVG, MAX, MIN
JOIN	连接	Combine data from multiple tables
Boolean	布尔	True/False logic
Logic gate	逻辑门	Electronic circuit implementing Boolean logic
AND gate	与门	Output true only if both inputs true
OR gate	或门	Output true if either input true
NOT gate	非门	Inverts the input
NAND gate	与非门	NOT-AND combination
NOR gate	或非门	NOT-OR combination
XOR gate	异或门	Output true if inputs are different
Truth table	真值表	Table showing all input/output combinations
Boolean expression	布尔表达式	Mathematical expression using AND, OR, NOT
Logic circuit	逻辑电路	Diagram of connected logic gates

SECTION 11: EXAM TECHNIQUE GUIDE

PAPER 1 EXAM TECHNIQUE (试卷 1 考试技巧)

TIME MANAGEMENT (时间管理)

Total time: 1 hour 45 minutes = 105 minutes

Total marks: 75 marks

Time per mark: 105 / 75 = 1.4 minutes per mark

Recommended approach:

- Section A (approx 30 marks): 30 x 1.4 = 42 minutes
- Section B (approx 30 marks): 30 x 1.4 = 42 minutes
- Section C (approx 15 marks): 15 x 1.4 = 21 minutes
- Review time: 105 - 105 = 0 minutes (!!!)

BETTER approach:

- Quick scan (2 min): Read all questions, note hard ones
- Section A (35 min): Answer all questions you can do quickly
- Section B (35 min): Structured questions -- show all working
- Section C (20 min): Extended answer -- plan before writing
- Review (10 min): Check answers, fill in gaps
- Hard questions (3 min): Attempt anything you skipped

ANSWERING DIFFERENT QUESTION TYPES (不同题型作答方法)

"State" or "Name" or "Give" (1 mark):

- Give a brief answer -- one word or short phrase

- No explanation needed
- Example: "State the number of bits in a byte" -> "8"

"Describe" (2-3 marks):

- Give details about HOW something works
- Use technical terms correctly
- Example: "Describe how parity checking works"

-> "An extra bit is added to make the total number of 1s odd (or even).

The receiver counts the 1s. If the parity is wrong, an error is detected."

"Explain" (2-4 marks):

- Give reasons WHY something happens or is done
- Link cause and effect
- Example: "Explain why cache memory improves performance"

-> "Cache is faster than RAM. The CPU checks cache first for data.

If found (cache hit), data is accessed quickly. This reduces waiting time."

"Compare" (3-4 marks):

- Give similarities AND differences
- Use connecting words: "whereas", "however", "both", "while"
- Example: "Compare RAM and ROM"

-> "RAM is volatile (loses data when power off) whereas ROM is non-volatile.

Both are types of primary memory. RAM is read/write while ROM is read-only."

"Evaluate" or "Discuss" (4-6 marks):

- Give advantages AND disadvantages
- Come to a conclusion or balanced judgement
- Use evidence or examples
- Example: "Evaluate the impact of AI on employment"

-> Positive: AI automates repetitive tasks, creates new jobs in tech.

-> Negative: Some jobs replaced, workers need retraining.

-> Conclusion: Overall positive if managed well, but social support needed.

COMMON ERRORS TO AVOID (常见错误)

- Not reading the question carefully (answer a different question!)
- Giving "Describe" answers when asked to "Explain"
- Not showing working in calculation questions
- Writing too little (1 line for a 4-mark question)
- Writing too much (wasting time on 1-mark questions)
- Forgetting units (bytes, Hz, bits, etc.)
- Confusing similar terms (RAM/ROM, compiler/interpreter, etc.)
- Not checking answers at the end

PAPER 2 EXAM TECHNIQUE (试卷 2 考试技巧)

TIME MANAGEMENT (时间管理)

Total time: 1 hour 45 minutes = 105 minutes

Total marks: 75 marks

Time per mark: 1.4 minutes per mark

Recommended approach:

- Quick scan (2 min): Look at scenario question, note topics
- Section A (20 min): Short answer questions (approx 25 marks)
- Section B (40 min): Code reading/writing (approx 35 marks)
- Section C (25 min): Scenario question (15 marks) -- PLAN FIRST!
- Review (18 min): Check code, check trace tables, fix errors

WRITING PSEUDOCODE (伪代码写作技巧)

- Use correct syntax from the syllabus
- Use meaningful variable names (e.g., TotalScore not T)
- Add comments using //
- Use indentation to show structure
- Always close your structures (ENDIF, ENDWHILE, NEXT, etc.)
- For input validation: use REPEAT...UNTIL or WHILE loops
- Initialise variables before using them
- Show your planning if the question asks for it

TRACE TABLES (跟踪表技巧)

- Set up columns for ALL variables mentioned
- Add a column for OUTPUT
- Work through ONE iteration at a time
- Be careful with loop conditions (check BEFORE or AFTER)
- Watch for off-by-one errors (starting at 0 vs 1)
- Don't rush -- trace tables are easy marks if careful

THE 15-MARK SCENARIO QUESTION (15 分情景题攻略)

STEP 1: READ (3 minutes)

- Read the scenario THREE times
- Underline key requirements
- Note: inputs needed, outputs required, processing required

STEP 2: PLAN (5 minutes)

- Draw a quick IPO chart (Input, Process, Output)
- Decide which subprograms to use
- Plan your variable names
- Note any validation requirements
- Planning is NOT wasted time -- it saves time!

STEP 3: CODE (10 minutes)

- Write structured pseudocode
- Use functions/procedures where appropriate
- Include input validation
- Add comments
- Leave space to add things you remember later

STEP 4: CHECK (7 minutes -- part of remaining review time)

- Does it meet ALL requirements?
- Are loops correctly structured?
- Have you closed all IF/LOOP structures?
- Are variable names meaningful?
- Did you include comments?

SQL WRITING TIPS (SQL 写作技巧)

- Always start with SELECT
- Use * only if ALL fields needed (otherwise name them)
- WHERE clause uses =, <>, <, >, <=, >=
- Use LIKE with wildcards (% for any characters, _ for single)
- ORDER BY comes after WHERE
- DESC for descending, ASC (or omit) for ascending
- Aggregate functions: COUNT(*), SUM(field), AVG(field)
- JOIN: SELECT * FROM Table1 INNER JOIN Table2 ON Table1.key = Table2.key

BOOLEAN LOGIC TIPS (布尔逻辑技巧)

- Draw truth tables systematically (list all combinations)
- For 2 inputs: 4 rows; 3 inputs: 8 rows; 4 inputs: 16 rows
- AND: only true if BOTH inputs true
- OR: true if EITHER input true
- NOT: inverts the input
- XOR: true if inputs are DIFFERENT
- Draw logic gates carefully -- the shape matters!

SECTION 12: "WHAT TO DO IF STUCK" EMERGENCY GUIDE

STUDENT EMERGENCY GUIDE (学生应急指南)

Print this guide and keep it in your revision folder.

Stick a copy inside your exam pencil case.

IF YOU DON'T UNDERSTAND A QUESTION:

10. READ IT AGAIN -- slowly, one word at a time
11. UNDERLINE the command word (State/Describe/Explain/Evaluate)
12. CIRCLE any technical terms in the question
13. ASK: "What topic is this about?" (binary? networks? loops?)
14. WRITE down everything you know about that topic
15. TRY to connect your knowledge to what the question asks
16. GUESS intelligently -- never leave blank!

IF YOU'RE STUCK ON A CALCULATION:

17. WRITE DOWN the formula you think you need
18. WRITE DOWN the values from the question
19. SHOW your working -- even if wrong, you get method marks
20. CHECK units (bits vs bytes, KB vs MB)
21. ESTIMATE: does your answer make sense?
22. If still stuck: MOVE ON, come back later

IF YOU'RE STUCK ON A CODING QUESTION:

23. READ the requirements again -- list what the program MUST do
24. WRITE comments first: // Get input, // Validate, // Calculate, // Output
25. FILL IN the code between your comments
26. USE a structure you know (IF, WHILE, FOR)
27. START SIMPLE -- get basic version working first
28. ADD validation and extras if time
29. If still stuck: write down your PLAN in words -- you get marks for design!

IF YOU'RE STUCK ON A TRACE TABLE:

30. WRITE DOWN the variable names as column headers
31. FOLLOW the code LINE BY LINE
32. UPDATE only the variables that change
33. CHECK loop conditions carefully
34. COUNT iterations -- how many times should it run?
35. If stuck: work through first 2 iterations carefully, then look for a pattern

IF YOU RUN OUT OF TIME:

36. DON'T PANIC -- this happens, you can still do well
37. QUICKLY scan remaining questions
38. ANSWER the highest-mark questions first
39. For short questions: write bullet points (may get partial marks)
40. For calculation questions: write the formula (method marks!)
41. NEVER leave a question completely blank

IF YOU FEEL PANICKED IN THE EXAM:

42. STOP writing
43. PUT DOWN your pen
44. TAKE 3 DEEP BREATHES (breathe in 4 seconds, out 4 seconds)
45. LOOK UP and around the room
46. DRINK some water
47. REMIND yourself: "I have prepared for this. I can do this."
48. START again with an easy question to build confidence

IF YOU FINISH EARLY:

49. DON'T leave early!
50. CHECK every answer
51. CHECK calculations (work backwards from your answer)
52. CHECK coding questions (are all loops closed?)
53. CHECK trace tables (re-do one iteration)
54. FILL IN any gaps you left
55. IMPROVE any short answers
56. Make sure your NAME is on every page!

REMEMBER:

>> A guess is better than a blank. (猜比空着好)

>> Show your working -- you get marks for method. (展示过程能得方法分)

>> Read the question twice before answering. (答题前读两遍题目)

>> Manage your time -- don't spend too long on one question. (管理好时间)

>> You know more than you think you do. (你掌握的知识比你以为的多)

>> Stay calm. Breathe. You've got this. (保持冷静。深呼吸。你能行。)

SECTION 13: SUMMARY AND CHECKLIST

END-OF-SEMESTER CHECKLIST FOR TEACHERS

[ ] All 5 mock exams administered under proper conditions

[ ] All mock papers marked with detailed feedback

[ ] All students have completed individual gap analysis

[ ] All students have 3 SMART targets recorded

[ ] All students have updated progress trackers

[ ] Class-level gap analysis completed (identify whole-class weak areas)

[ ] Targeted revision sessions delivered for identified gaps

[ ] Revision folders complete and organised

[ ] Exam technique guide distributed to all students

[ ] "What to do if stuck" guide distributed to all students

[ ] Bilingual glossary provided to all students

[ ] Exam day logistics confirmed (room, time, materials)

[ ] Parents informed of exam dates and preparation expectations

[ ] Students emotionally prepared and confident

[ ] Celebration of hard work planned for end of course

END-OF-SEMESTER CHECKLIST FOR STUDENTS

[ ] I have completed all 5 mock exams

[ ] I know my marks for all 5 mocks

[ ] I have identified my 3 weakest topics

[ ] I have revised my 3 weakest topics

[ ] I have a revision plan for the exam period

[ ] I can answer questions on all 10 syllabus topics

[ ] I can write pseudocode for common algorithms

[ ] I can complete trace tables accurately

[ ] I can write SQL queries

[ ] I can draw and interpret logic circuits

[ ] I understand all command words (State, Describe, Explain, Evaluate)

[ ] I can manage my time in a 105-minute exam

[ ] I know what to do if I get stuck

[ ] I have all necessary equipment for the exam

[ ] I feel confident and ready!

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